Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z4,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z4 ZL: $\frac{Z_1 Z_4 Z_L g_m}{Z_1 Z_4 g_m + 2 Z_1 Z_L g_m + Z_4 + 2 Z_L}$

$$H(z) = \frac{Z_1 Z_4 Z_L g_m}{Z_1 Z_4 g_m + 2 Z_1 Z_L g_m + Z_4 + 2 Z_L}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ R_4,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_4 g_m}{C_1 C_L R_4 s^2 + 2 g_m + s \left(2 C_1 + C_L R_4 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}\sqrt{R_4}\sqrt{g_m}}{2C_1 + C_L R_4 g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_4}}$
bandwidth: $\frac{2C_1 + C_L R_4 g_m}{C_1 C_L R_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.2 LP-2 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ R_4,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_4 R_L g_m}{C_1 C_L R_4 R_L s^2 + R_4 g_m + 2 R_L g_m + s \left(C_1 R_4 + 2 C_1 R_L + C_L R_4 R_L g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}\sqrt{g_m}\sqrt{R_4 + 2R_L}}{C_1 R_4 + 2C_1 R_L + C_L R_4 R_L g_m}$
wo: $\frac{\sqrt{R_4 g_m + 2R_L g_m}}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}}$
bandwidth: $\frac{\sqrt{R_4 g_m + 2R_L g_m}(C_1 R_4 + 2C_1 R_L + C_L R_4 R_L g_m)}{C_1 C_L R_4 R_L \sqrt{g_m}\sqrt{R_4 + 2R_L}}$
K-LP: $\frac{R_4 R_L}{R_4 + 2R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.3 LP-3 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ R_L\right)$

$$H(s) = \frac{R_L g_m}{2C_1C_4R_Ls^2 + g_m + s\left(C_1 + 2C_4R_Lg_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_L}\sqrt{g_m}}{C_1+2C_4R_Lg_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_L}}$
bandwidth: $\frac{C_1+2C_4R_Lg_m}{2C_1C_4R_L}$
K-LP: R_L
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.4 LP-4 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \frac{R_L}{C_LR_Ls+1}\right)$

$$H(s) = \frac{R_L g_m}{g_m + s^2\left(2C_1C_4R_L + C_1C_LR_L\right) + s\left(C_1 + 2C_4R_Lg_m + C_LR_Lg_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_L}\sqrt{g_m}\sqrt{2C_4+C_L}}{C_1+2C_4R_Lg_m+C_LR_Lg_m}$
wo: $\frac{\sqrt{g_m}}{\sqrt{2C_1C_4R_L+C_1C_LR_L}}$
bandwidth: $\frac{C_1+2C_4R_Lg_m+C_LR_Lg_m}{\sqrt{C_1}\sqrt{R_L}\sqrt{2C_4+C_L}\sqrt{2C_1C_4R_L+C_1C_LR_L}}$
K-LP: R_L
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.5 LP-5 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4}{C_4R_4s+1},\ \infty,\ R_L\right)$

$$H(s) = \frac{R_4R_L g_m}{2C_1C_4R_4R_Ls^2 + R_4g_m + 2R_Lg_m + s\left(C_1R_4 + 2C_1R_L + 2C_4R_4R_Lg_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{R_L}\sqrt{g_m}\sqrt{R_4+2R_L}}{C_1R_4+2C_1R_L+2C_4R_4R_Lg_m}$
wo: $\frac{\sqrt{2}\sqrt{R_4g_m+2R_Lg_m}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{R_L}}$
bandwidth: $\frac{\sqrt{R_4g_m+2R_Lg_m}(C_1R_4+2C_1R_L+2C_4R_4R_Lg_m)}{2C_1C_4R_4R_L\sqrt{g_m}\sqrt{R_4+2R_L}}$
K-LP: $\frac{R_4R_L}{R_4+2R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.6 LP-6 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_4 g_m}{2g_m + s^2\left(2C_1C_4R_4 + C_1C_LR_4\right) + s\left(2C_1 + 2C_4R_4g_m + C_LR_4g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{R_4}\sqrt{g_m}\sqrt{2C_4+C_L}}{2C_1+2C_4R_4g_m+C_LR_4g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{2C_1C_4R_4+C_1C_LR_4}}$
bandwidth: $\frac{2C_1+2C_4R_4g_m+C_LR_4g_m}{\sqrt{C_1}\sqrt{R_4}\sqrt{2C_4+C_L}\sqrt{2C_1C_4R_4+C_1C_LR_4}}$

K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.7 LP-7 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_4 R_L g_m}{R_4 g_m + 2 R_L g_m + s^2 \left(2 C_1 C_4 R_4 R_L + C_1 C_L R_4 R_L\right) + s \left(C_1 R_4 + 2 C_1 R_L + 2 C_4 R_4 R_L g_m + C_L R_4 R_L g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_4}\sqrt{R_L}\sqrt{g_m}\sqrt{2C_4+C_L}\sqrt{R_4+2R_L}}{C_1 R_4+2C_1 R_L+2C_4 R_4 R_L g_m+C_L R_4 R_L g_m}$
wo: $\frac{\sqrt{R_4 g_m+2R_L g_m}}{\sqrt{2C_1 C_4 R_4 R_L+C_1 C_L R_4 R_L}}$
bandwidth: $\frac{\sqrt{R_4 g_m+2R_L g_m}(C_1 R_4+2C_1 R_L+2C_4 R_4 R_L g_m+C_L R_4 R_L g_m)}{\sqrt{C_1}\sqrt{R_4}\sqrt{R_L}\sqrt{g_m}\sqrt{2C_4+C_L}\sqrt{R_4+2R_L}\sqrt{2C_1 C_4 R_4 R_L+C_1 C_L R_4 R_L}}$
K-LP: $\frac{R_4 R_L}{R_4+2R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.8 LP-8 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ R_4,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_4 g_m}{C_1 C_L R_1 R_4 s^2 + 2 R_1 g_m + s \left(2 C_1 R_1 + C_L R_1 R_4 g_m + C_L R_4\right) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1 g_m+1}}{2C_1 R_1+C_L R_1 R_4 g_m+C_L R_4}$
wo: $\frac{\sqrt{2R_1 g_m+2}}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_1 g_m+2}(2C_1 R_1+C_L R_1 R_4 g_m+C_L R_4)}{2C_1 C_L R_1 R_4\sqrt{R_1 g_m+1}}$
K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m+2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.9 LP-9 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ R_4,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_1 R_4 R_L g_m}{C_1 C_L R_1 R_4 R_L s^2 + R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L + s \left(C_1 R_1 R_4 + 2 C_1 R_1 R_L + C_L R_1 R_4 R_L g_m + C_L R_4 R_L\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_4}\sqrt{R_L}\sqrt{R_1 R_4 g_m+2R_1 R_L g_m+R_4+2R_L}}{C_1 R_1 R_4+2C_1 R_1 R_L+C_L R_1 R_4 R_L g_m+C_L R_4 R_L}$
wo: $\frac{\sqrt{R_1 R_4 g_m+2R_1 R_L g_m+R_4+2R_L}}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_4}\sqrt{R_L}}$
bandwidth: $\frac{C_1 R_1 R_4+2C_1 R_1 R_L+C_L R_1 R_4 R_L g_m+C_L R_4 R_L}{C_1 C_L R_1 R_4 R_L}$
K-LP: $\frac{R_1 R_4 R_L g_m}{R_1 R_4 g_m+2R_1 R_L g_m+R_4+2R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.10 LP-10 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ \frac{1}{C_4s},\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1R_Lg_m}{2C_1C_4R_1R_Ls^2 + R_1g_m + s\left(C_1R_1 + 2C_4R_1R_Lg_m + 2C_4R_L\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_L}\sqrt{R_1g_m+1}}{C_1R_1+2C_4R_1R_Lg_m+2C_4R_L}$
wo: $\frac{\sqrt{2}\sqrt{R_1g_m+1}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_L}}$
bandwidth: $\frac{C_1R_1+2C_4R_1R_Lg_m+2C_4R_L}{2C_1C_4R_1R_L}$
K-LP: $\frac{R_1R_Lg_m}{R_1g_m+1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.11 LP-11 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ \frac{1}{C_4s},\ \infty,\ \frac{R_L}{C_LR_Ls+1}\right)$

$$H(s) = \frac{R_1R_Lg_m}{R_1g_m + s^2\left(2C_1C_4R_1R_L + C_1C_LR_1R_L\right) + s\left(C_1R_1 + 2C_4R_1R_Lg_m + 2C_4R_L + C_LR_1R_Lg_m + C_LR_L\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_1}\sqrt{R_L}\sqrt{2C_4+C_L}\sqrt{R_1g_m+1}}{C_1R_1+2C_4R_1R_Lg_m+2C_4R_L+C_LR_1R_Lg_m+C_LR_L}$
wo: $\frac{\sqrt{R_1g_m+1}}{\sqrt{2C_1C_4R_1R_L+C_1C_LR_1R_L}}$
bandwidth: $\frac{C_1R_1+2C_4R_1R_Lg_m+2C_4R_L+C_LR_1R_Lg_m+C_LR_L}{\sqrt{C_1}\sqrt{R_1}\sqrt{R_L}\sqrt{2C_4+C_L}\sqrt{2C_1C_4R_1R_L+C_1C_LR_1R_L}}$
K-LP: $\frac{R_1R_Lg_m}{R_1g_m+1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.12 LP-12 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ \frac{R_4}{C_4R_4s+1},\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1R_4R_Lg_m}{2C_1C_4R_1R_4R_Ls^2 + R_1R_4g_m + 2R_1R_Lg_m + R_4 + 2R_L + s\left(C_1R_1R_4 + 2C_1R_1R_L + 2C_4R_1R_4R_Lg_m + 2C_4R_4R_L\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}\sqrt{R_L}\sqrt{R_1R_4g_m+2R_1R_Lg_m+R_4+2R_L}}{C_1R_1R_4+2C_1R_1R_L+2C_4R_1R_4R_Lg_m+2C_4R_4R_L}$
wo: $\frac{\sqrt{2}\sqrt{R_1R_4g_m+2R_1R_Lg_m+R_4+2R_L}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}\sqrt{R_L}}$
bandwidth: $\frac{C_1R_1R_4+2C_1R_1R_L+2C_4R_1R_4R_Lg_m+2C_4R_4R_L}{2C_1C_4R_1R_4R_L}$
K-LP: $\frac{R_1R_4R_Lg_m}{R_1R_4g_m+2R_1R_Lg_m+R_4+2R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.13 LP-13 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_1R_4g_m}{2R_1g_m + s^2\left(2C_1C_4R_1R_4 + C_1C_LR_1R_4\right) + s\left(2C_1R_1 + 2C_4R_1R_4g_m + 2C_4R_4 + C_LR_1R_4g_m + C_LR_4\right) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{R_1}\sqrt{R_4}\sqrt{2C_4+C_L}\sqrt{R_1g_m+1}}{2C_1R_1+2C_4R_1R_4g_m+2C_4R_4+C_LR_1R_4g_m+C_LR_4}$
wo: $\frac{\sqrt{2R_1g_m+2}}{\sqrt{2C_1C_4R_1R_4+C_1C_LR_1R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_1g_m+2}\left(2C_1R_1+2C_4R_1R_4g_m+2C_4R_4+C_LR_1R_4g_m+C_LR_4\right)}{2\sqrt{C_1}\sqrt{R_1}\sqrt{R_4}\sqrt{2C_4+C_L}\sqrt{R_1g_m+1}\sqrt{2C_1C_4R_1R_4+C_1C_LR_1R_4}}$

K-LP: $\frac{R_1 R_4 g_m}{2 R_1 g_m + 2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.14 LP-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s + 1},\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_1 R_4 R_L g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L + s^2 \left(2 C_1 C_4 R_1 R_4 R_L + C_1 C_L R_1 R_4 R_L\right) + s \left(C_1 R_1 R_4 + 2 C_1 R_1 R_L + 2 C_4 R_1 R_4 R_L g_m + 2 C_4 R_4 R_L + C_L R_1 R_4 R_L g_m + C_L R_4 R_L\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_1}\sqrt{R_4}\sqrt{R_L}\sqrt{2C_4 + C_L}\sqrt{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}}{C_1 R_1 R_4 + 2 C_1 R_1 R_L + 2 C_4 R_1 R_4 R_L g_m + 2 C_4 R_4 R_L + C_L R_1 R_4 R_L g_m + C_L R_4 R_L}$
wo: $\frac{\sqrt{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}}{\sqrt{2 C_1 C_4 R_1 R_4 R_L + C_1 C_L R_1 R_4 R_L}}$
bandwidth: $\frac{C_1 R_1 R_4 + 2 C_1 R_1 R_L + 2 C_4 R_1 R_4 R_L g_m + 2 C_4 R_4 R_L + C_L R_1 R_4 R_L g_m + C_L R_4 R_L}{\sqrt{C_1}\sqrt{R_1}\sqrt{R_4}\sqrt{R_L}\sqrt{2C_4 + C_L}\sqrt{2 C_1 C_4 R_1 R_4 R_L + C_1 C_L R_1 R_4 R_L}}$
K-LP: $\frac{R_1 R_4 R_L g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s + 1},\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_1 R_4 R_L g_m s + R_1 R_4 g_m}{2 R_1 g_m + s^2 \left(2 C_4 C_L R_1 R_4 R_L g_m + 2 C_4 C_L R_4 R_L\right) + s \left(2 C_4 R_1 R_4 g_m + 2 C_4 R_4 + C_L R_1 R_4 g_m + 2 C_L R_1 R_L g_m + C_L R_4 + 2 C_L R_L\right) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}}{2 C_4 R_4 + C_L R_4 + 2 C_L R_L}$
wo: $\frac{1}{\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}}$
bandwidth: $\frac{2 C_4 R_4 + C_L R_4 + 2 C_L R_L}{2 C_4 C_L R_4 R_L}$
K-LP: $\frac{R_1 R_4 g_m}{2 R_1 g_m + 2}$
K-HP: 0
K-BP: $\frac{C_L R_1 R_4 R_L g_m}{2 C_4 R_1 R_4 g_m + 2 C_4 R_4 + C_L R_1 R_4 g_m + 2 C_L R_1 R_L g_m + C_L R_4 + 2 C_L R_L}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1,\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{C_4 R_1 R_4 R_L g_m s + R_1 R_L g_m}{R_1 g_m + s^2 \left(C_4 C_L R_1 R_4 R_L g_m + C_4 C_L R_4 R_L\right) + s \left(C_4 R_1 R_4 g_m + 2 C_4 R_1 R_L g_m + C_4 R_4 + 2 C_4 R_L + C_L R_1 R_L g_m + C_L R_L\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}}{C_4 R_4 + 2 C_4 R_L + C_L R_L}$
wo: $\frac{1}{\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}}$
bandwidth: $\frac{C_4 R_4 + 2 C_4 R_L + C_L R_L}{C_4 C_L R_4 R_L}$
K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_4 R_1 R_4 R_L g_m}{C_4 R_1 R_4 g_m + 2 C_4 R_1 R_L g_m + C_4 R_4 + 2 C_4 R_L + C_L R_1 R_L g_m + C_L R_L}$
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ R_4,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_4 R_L g_m s + R_4 g_m}{2g_m + s^2 \left(C_1 C_L R_4 + 2C_1 C_L R_L\right) + s \left(2C_1 + C_L R_4 g_m + 2C_L R_L g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}\sqrt{g_m}\sqrt{R_4+2R_L}}{2C_1+C_L R_4 g_m+2C_L R_L g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1 C_L R_4+2C_1 C_L R_L}}$
bandwidth: $\frac{2C_1+C_L R_4 g_m+2C_L R_L g_m}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_4+2R_L}\sqrt{C_1 C_L R_4+2C_1 C_L R_L}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_L R_4 R_L g_m}{2C_1+C_L R_4 g_m+2C_L R_L g_m}$
Qz: None
Wz: None

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ R_L\right)$

$$H(s) = \frac{C_4 R_4 R_L g_m s + R_L g_m}{g_m + s^2 \left(C_1 C_4 R_4 + 2C_1 C_4 R_L\right) + s \left(C_1 + C_4 R_4 g_m + 2C_4 R_L g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_4}\sqrt{g_m}\sqrt{R_4+2R_L}}{C_1+C_4 R_4 g_m+2C_4 R_L g_m}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_1 C_4 R_4+2C_1 C_4 R_L}}$
bandwidth: $\frac{C_1+C_4 R_4 g_m+2C_4 R_L g_m}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_4+2R_L}\sqrt{C_1 C_4 R_4+2C_1 C_4 R_L}}$
K-LP: R_L
K-HP: 0
K-BP: $\frac{C_4 R_4 R_L g_m}{C_1+C_4 R_4 g_m+2C_4 R_L g_m}$
Qz: None
Wz: None

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ R_4,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_1 R_4 R_L g_m s + R_1 R_4 g_m}{2R_1 g_m + s^2 \left(C_1 C_L R_1 R_4 + 2C_1 C_L R_1 R_L\right) + s \left(2C_1 R_1 + C_L R_1 R_4 g_m + 2C_L R_1 R_L g_m + C_L R_4 + 2C_L R_L\right) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_4+2R_L}\sqrt{R_1 g_m+1}}{2C_1 R_1+C_L R_1 R_4 g_m+2C_L R_1 R_L g_m+C_L R_4+2C_L R_L}$
wo: $\frac{\sqrt{2R_1 g_m+2}}{\sqrt{C_1 C_L R_1 R_4+2C_1 C_L R_1 R_L}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_1 g_m+2}(2C_1 R_1+C_L R_1 R_4 g_m+2C_L R_1 R_L g_m+C_L R_4+2C_L R_L)}{2\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_4+2R_L}\sqrt{R_1 g_m+1}\sqrt{C_1 C_L R_1 R_4+2C_1 C_L R_1 R_L}}$
K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m+2}$
K-HP: 0
K-BP: $\frac{C_L R_1 R_4 R_L g_m}{2C_1 R_1+C_L R_1 R_4 g_m+2C_L R_1 R_L g_m+C_L R_4+2C_L R_L}$
Qz: None
Wz: None

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ R_L\right)$

$$H(s) = \frac{C_4 R_1 R_4 R_L g_m s + R_1 R_L g_m}{R_1 g_m + s^2 \left(C_1 C_4 R_1 R_4 + 2C_1 C_4 R_1 R_L\right) + s \left(C_1 R_1 + C_4 R_1 R_4 g_m + 2C_4 R_1 R_L g_m + C_4 R_4 + 2C_4 R_L\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4+2R_L}\sqrt{R_1 g_m+1}}{C_1 R_1+C_4 R_1 R_4 g_m+2C_4 R_1 R_L g_m+C_4 R_4+2C_4 R_L}$
wo: $\frac{\sqrt{R_1 g_m+1}}{\sqrt{C_1 C_4 R_1 R_4+2C_1 C_4 R_1 R_L}}$
bandwidth: $\frac{C_1 R_1+C_4 R_1 R_4 g_m+2C_4 R_1 R_L g_m+C_4 R_4+2C_4 R_L}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4+2R_L}\sqrt{C_1 C_4 R_1 R_4+2C_1 C_4 R_1 R_L}}$

K-LP: $\frac{R_1 R_L g_m}{R_1 g_m+1}$
K-HP: 0
K-BP: $\frac{C_4 R_1 R_4 R_L g_m}{C_1 R_1+C_4 R_1 R_4 g_m+2C_4 R_1 R_L g_m+C_4 R_4+2C_4 R_L}$
Qz: None
Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ R_4,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_1 R_1 R_4 g_m s + R_4 g_m}{2g_m + s^2\left(C_1 C_L R_1 R_4 g_m + C_1 C_L R_4\right) + s\left(2C_1 R_1 g_m + 2C_1 + C_L R_4 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}g_m\sqrt{\frac{g_m}{R_1 g_m+1}}+\sqrt{2}\sqrt{C_1}\sqrt{C_L}\sqrt{R_4}\sqrt{\frac{g_m}{R_1 g_m+1}}}{2C_1 R_1 g_m+2C_1+C_L R_4 g_m}$
wo: $\sqrt{2}\sqrt{\frac{g_m}{C_1 C_L R_1 R_4 g_m+C_1 C_L R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{C_1 C_L R_1 R_4 g_m+C_1 C_L R_4}}\left(2C_1 R_1 g_m+2C_1+C_L R_4 g_m\right)}{\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}g_m\sqrt{\frac{g_m}{R_1 g_m+1}}+\sqrt{2}\sqrt{C_1}\sqrt{C_L}\sqrt{R_4}\sqrt{\frac{g_m}{R_1 g_m+1}}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_4 g_m}{2C_1 R_1 g_m+2C_1+C_L R_4 g_m}$
Qz: None
Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ R_4,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{C_1 R_1 R_4 R_L g_m s + R_4 R_L g_m}{R_4 g_m + 2R_L g_m + s^2\left(C_1 C_L R_1 R_4 R_L g_m + C_1 C_L R_4 R_L\right) + s\left(C_1 R_1 R_4 g_m + 2C_1 R_1 R_L g_m + C_1 R_4 + 2C_1 R_L + C_L R_4 R_L g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}\sqrt{R_L}g_m\sqrt{\frac{g_m}{R_1 g_m+1}}\sqrt{R_4+2R_L}+\sqrt{C_1}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{g_m}{R_1 g_m+1}}\sqrt{R_4+2R_L}}{C_1 R_1 R_4 g_m+2C_1 R_1 R_L g_m+C_1 R_4+2C_1 R_L+C_L R_4 R_L g_m}$
wo: $\sqrt{\frac{R_4 g_m+2R_L g_m}{C_1 C_L R_1 R_4 R_L g_m+C_1 C_L R_4 R_L}}$
bandwidth: $\frac{\sqrt{\frac{R_4 g_m+2R_L g_m}{C_1 C_L R_1 R_4 R_L g_m+C_1 C_L R_4 R_L}}\left(C_1 R_1 R_4 g_m+2C_1 R_1 R_L g_m+C_1 R_4+2C_1 R_L+C_L R_4 R_L g_m\right)}{\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}\sqrt{R_L}g_m\sqrt{\frac{g_m}{R_1 g_m+1}}\sqrt{R_4+2R_L}+\sqrt{C_1}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{g_m}{R_1 g_m+1}}\sqrt{R_4+2R_L}}$
K-LP: $\frac{R_4 R_L}{R_4+2R_L}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_4 R_L g_m}{C_1 R_1 R_4 g_m+2C_1 R_1 R_L g_m+C_1 R_4+2C_1 R_L+C_L R_4 R_L g_m}$
Qz: None
Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ R_L\right)$

$$H(s) = \frac{C_1 R_1 R_L g_m s + R_L g_m}{g_m + s^2\left(2C_1 C_4 R_1 R_L g_m + 2C_1 C_4 R_L\right) + s\left(C_1 R_1 g_m + C_1 + 2C_4 R_L g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_L}g_m\sqrt{\frac{g_m}{R_1 g_m+1}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_L}\sqrt{\frac{g_m}{R_1 g_m+1}}}{C_1 R_1 g_m+C_1+2C_4 R_L g_m}$
wo: $\sqrt{\frac{g_m}{2C_1 C_4 R_1 R_L g_m+2C_1 C_4 R_L}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{2C_1 C_4 R_1 R_L g_m+2C_1 C_4 R_L}}\left(C_1 R_1 g_m+C_1+2C_4 R_L g_m\right)}{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_L}g_m\sqrt{\frac{g_m}{R_1 g_m+1}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_L}\sqrt{\frac{g_m}{R_1 g_m+1}}}$
K-LP: R_L
K-HP: 0
K-BP: $\frac{C_1 R_1 R_L g_m}{C_1 R_1 g_m+C_1+2C_4 R_L g_m}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{C_1 R_1 R_L g_m s + R_L g_m}{g_m + s^2 \left(2C_1C_4R_1R_Lg_m + 2C_1C_4R_L + C_1C_LR_1R_Lg_m + C_1C_LR_L\right) + s\left(C_1R_1g_m + C_1 + 2C_4R_Lg_m + C_LR_Lg_m\right)}$$

Parameters:

Q: $\frac{2\sqrt{C_1}C_4R_1\sqrt{R_L}g_m\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+2\sqrt{C_1}C_4\sqrt{R_L}\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+\sqrt{C_1}C_LR_1\sqrt{R_L}g_m\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+\sqrt{C_1}C_L\sqrt{R_L}\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}}{C_1R_1g_m+C_1+2C_4R_Lg_m+C_LR_Lg_m}$
wo: $\sqrt{\frac{g_m}{2C_1C_4R_1R_Lg_m+2C_1C_4R_L+C_1C_LR_1R_Lg_m+C_1C_LR_L}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{2C_1C_4R_1R_Lg_m+2C_1C_4R_L+C_1C_LR_1R_Lg_m+C_1C_LR_L}}\left(C_1R_1g_m+C_1+2C_4R_Lg_m+C_LR_Lg_m\right)}{2\sqrt{C_1}C_4R_1\sqrt{R_L}g_m\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+2\sqrt{C_1}C_4\sqrt{R_L}\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+\sqrt{C_1}C_LR_1\sqrt{R_L}g_m\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+\sqrt{C_1}C_L\sqrt{R_L}\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}}$
K-LP: R_L
K-HP: 0
K-BP: $\frac{C_1R_1R_Lg_m}{C_1R_1g_m+C_1+2C_4R_Lg_m+C_LR_Lg_m}$
Qz: None
Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ R_L\right)$

$$H(s) = \frac{C_1 R_1 R_4 R_L g_m s + R_4 R_L g_m}{R_4 g_m + 2R_L g_m + s^2 \left(2C_1C_4R_1R_4R_Lg_m + 2C_1C_4R_4R_L\right) + s\left(C_1R_1R_4g_m + 2C_1R_1R_Lg_m + C_1R_4 + 2C_1R_L + 2C_4R_4R_Lg_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{R_L}g_m\sqrt{\frac{g_m}{R_1g_m+1}}\sqrt{R_4+2R_L}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{g_m}{R_1g_m+1}}\sqrt{R_4+2R_L}}{C_1R_1R_4g_m+2C_1R_1R_Lg_m+C_1R_4+2C_1R_L+2C_4R_4R_Lg_m}$
wo: $\sqrt{\frac{R_4g_m+2R_Lg_m}{2C_1C_4R_1R_4R_Lg_m+2C_1C_4R_4R_L}}$
bandwidth: $\frac{\sqrt{\frac{R_4g_m+2R_Lg_m}{2C_1C_4R_1R_4R_Lg_m+2C_1C_4R_4R_L}}\left(C_1R_1R_4g_m+2C_1R_1R_Lg_m+C_1R_4+2C_1R_L+2C_4R_4R_Lg_m\right)}{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{R_L}g_m\sqrt{\frac{g_m}{R_1g_m+1}}\sqrt{R_4+2R_L}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{g_m}{R_1g_m+1}}\sqrt{R_4+2R_L}}$
K-LP: $\frac{R_4R_L}{R_4+2R_L}$
K-HP: 0
K-BP: $\frac{C_1R_1R_4R_Lg_m}{C_1R_1R_4g_m+2C_1R_1R_Lg_m+C_1R_4+2C_1R_L+2C_4R_4R_Lg_m}$
Qz: None
Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_1 R_1 R_4 g_m s + R_4 g_m}{2g_m + s^2 \left(2C_1C_4R_1R_4g_m + 2C_1C_4R_4 + C_1C_LR_1R_4g_m + C_1C_LR_4\right) + s\left(2C_1R_1g_m + 2C_1 + 2C_4R_4g_m + C_LR_4g_m\right)}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{R_4}g_m\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+2\sqrt{2}\sqrt{C_1}C_4\sqrt{R_4}\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+\sqrt{2}\sqrt{C_1}C_LR_1\sqrt{R_4}g_m\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+\sqrt{2}\sqrt{C_1}C_L\sqrt{R_4}\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}}{2C_1R_1g_m+2C_1+2C_4R_4g_m+C_LR_4g_m}$
wo: $\sqrt{2}\sqrt{\frac{g_m}{2C_1C_4R_1R_4g_m+2C_1C_4R_4+C_1C_LR_1R_4g_m+C_1C_LR_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_4R_1R_4g_m+2C_1C_4R_4+C_1C_LR_1R_4g_m+C_1C_LR_4}}\left(2C_1R_1g_m+2C_1+2C_4R_4g_m+C_LR_4g_m\right)}{2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{R_4}g_m\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+2\sqrt{2}\sqrt{C_1}C_4\sqrt{R_4}\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+\sqrt{2}\sqrt{C_1}C_LR_1\sqrt{R_4}g_m\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}+\sqrt{2}\sqrt{C_1}C_L\sqrt{R_4}\sqrt{\frac{g_m}{2C_4R_1g_m+2C_4+C_LR_1g_m+C_L}}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_1R_1R_4g_m}{2C_1R_1g_m+2C_1+2C_4R_4g_m+C_LR_4g_m}$
Qz: None
Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s + 1},\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{C_1 R_1 R_4 R_L g_m s + R_4 R_L g_m}{R_4 g_m + 2R_L g_m + s^2 \left(2C_1 C_4 R_1 R_4 R_L g_m + 2C_1 C_4 R_4 R_L + C_1 C_L R_1 R_4 R_L g_m + C_1 C_L R_4 R_L\right) + s \left(C_1 R_1 R_4 g_m + 2C_1 R_1 R_L g_m + C_1 R_4 + 2C_1 R_L + 2C_4 R_4 R_L g_m + C_L R_4 R_L g_m\right)}$$

Parameters:

Q: $\frac{2\sqrt{C_1} C_4 R_1 \sqrt{R_4} \sqrt{R_L} g_m \sqrt{\frac{g_m}{2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L}} \sqrt{R_4 + 2R_L} + 2\sqrt{C_1} C_4 \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{g_m}{2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L}} \sqrt{R_4 + 2R_L} + \sqrt{C_1} C_L R_1 \sqrt{R_4} \sqrt{R_L} g_m \sqrt{\frac{g_m}{2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L}} \sqrt{R_4 + 2R_L} + \sqrt{C_1} C_L \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{g_m}{2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L}} \sqrt{R_4 + 2R_L}}{C_1 R_1 R_4 g_m + 2C_1 R_1 R_L g_m + C_1 R_4 + 2C_1 R_L + 2C_4 R_4 R_L g_m + C_L R_4 R_L g_m}$

wo: $\sqrt{\frac{R_4 g_m + 2R_L g_m}{2C_1 C_4 R_1 R_4 R_L g_m + 2C_1 C_4 R_4 R_L + C_1 C_L R_1 R_4 R_L g_m + C_1 C_L R_4 R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_4 g_m + 2R_L g_m}{2C_1 C_4 R_1 R_4 R_L g_m + 2C_1 C_4 R_4 R_L + C_1 C_L R_1 R_4 R_L g_m + C_1 C_L R_4 R_L}} \left(C_1 R_1 R_4 g_m + 2C_1 R_1 R_L g_m + C_1 R_4 + 2C_1 R_L + 2C_4 R_4 R_L g_m + C_L R_4 R_L g_m\right)}{2\sqrt{C_1} C_4 R_1 \sqrt{R_4} \sqrt{R_L} g_m \sqrt{\frac{g_m}{2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L}} \sqrt{R_4 + 2R_L} + 2\sqrt{C_1} C_4 \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{g_m}{2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L}} \sqrt{R_4 + 2R_L} + \sqrt{C_1} C_L R_1 \sqrt{R_4} \sqrt{R_L} g_m \sqrt{\frac{g_m}{2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L}} \sqrt{R_4 + 2R_L} + \sqrt{C_1} C_L \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{g_m}{2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L}} \sqrt{R_4 + 2R_L}}$

K-LP: $\frac{R_4 R_L}{R_4 + 2R_L}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_4 R_L g_m}{C_1 R_1 R_4 g_m + 2C_1 R_1 R_L g_m + C_1 R_4 + 2C_1 R_L + 2C_4 R_4 R_L g_m + C_L R_4 R_L g_m}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1,\ \infty,\ \infty,\ R_4,\ \infty,\ R_L)$

$$H(s) = \frac{R_1 R_4 R_L g_m}{R_1 R_4 g_m + 2R_1 R_L g_m + R_4 + 2R_L}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1,\ \infty,\ \infty,\ R_4,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_4 g_m}{2R_1 g_m + s \left(C_L R_1 R_4 g_m + C_L R_4\right) + 2}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1,\ \infty,\ \infty,\ R_4,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_1 R_4 R_L g_m}{R_1 R_4 g_m + 2R_1 R_L g_m + R_4 + 2R_L + s \left(C_L R_1 R_4 R_L g_m + C_L R_4 R_L\right)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1,\ \infty,\ \infty,\ R_4,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_1 R_4 R_L g_m s + R_1 R_4 g_m}{2R_1 g_m + s \left(C_L R_1 R_4 g_m + 2C_L R_1 R_L g_m + C_L R_4 + 2C_L R_L\right) + 2}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1 R_L g_m}{R_1 g_m + s \left(2C_4 R_1 R_L g_m + 2C_4 R_L\right) + 1}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m}{s \left(2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L\right)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_1 R_L g_m}{R_1 g_m + s\left(2C_4 R_1 R_L g_m + 2C_4 R_L + C_L R_1 R_L g_m + C_L R_L\right) + 1}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_1 R_L g_m s + R_1 g_m}{s^2\left(2C_4 C_L R_1 R_L g_m + 2C_4 C_L R_L\right) + s\left(2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L\right)}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1 R_4 R_L g_m}{R_1 R_4 g_m + 2R_1 R_L g_m + R_4 + 2R_L + s\left(2C_4 R_1 R_4 R_L g_m + 2C_4 R_4 R_L\right)}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_4 g_m}{2R_1 g_m + s\left(2C_4 R_1 R_4 g_m + 2C_4 R_4 + C_L R_1 R_4 g_m + C_L R_4\right) + 2}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_1 R_4 R_L g_m}{R_1 R_4 g_m + 2R_1 R_L g_m + R_4 + 2R_L + s\left(2C_4 R_1 R_4 R_L g_m + 2C_4 R_4 R_L + C_L R_1 R_4 R_L g_m + C_L R_4 R_L\right)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(R_1,\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ R_L\right)$

$$H(s) = \frac{C_4 R_1 R_4 R_L g_m s + R_1 R_L g_m}{R_1 g_m + s\left(C_4 R_1 R_4 g_m + 2C_4 R_1 R_L g_m + C_4 R_4 + 2C_4 R_L\right) + 1}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(R_1,\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_4 R_1 R_4 g_m s + R_1 g_m}{s^2\left(C_4 C_L R_1 R_4 g_m + C_4 C_L R_4\right) + s\left(2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L\right)}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(R_1,\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_4 C_L R_1 R_4 R_L g_m s^2 + R_1 g_m + s\left(C_4 R_1 R_4 g_m + C_L R_1 R_L g_m\right)}{s^2\left(C_4 C_L R_1 R_4 g_m + 2C_4 C_L R_1 R_L g_m + C_4 C_L R_4 + 2C_4 C_L R_L\right) + s\left(2C_4 R_1 g_m + 2C_4 + C_L R_1 g_m + C_L\right)}$$

9.15 X-INVALID-ORDER-15 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ R_4,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_4 R_L g_m}{R_4 g_m + 2R_L g_m + s\left(C_1 R_4 + 2C_1 R_L\right)}$$

9.16 X-INVALID-ORDER-16 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m}{s^2\left(2C_1 C_4 + C_1 C_L\right) + s\left(2C_4 g_m + C_L g_m\right)}$$

9.17 X-INVALID-ORDER-17 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_L g_m s + g_m}{2C_1C_4C_LR_Ls^3 + s^2\left(2C_1C_4 + C_1C_L + 2C_4C_LR_Lg_m\right) + s\left(2C_4g_m + C_Lg_m\right)}$$

9.18 X-INVALID-ORDER-18 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_4 R_L g_m s + R_4 g_m}{2C_1C_4C_LR_4R_Ls^3 + 2g_m + s^2\left(2C_1C_4R_4 + C_1C_LR_4 + 2C_1C_LR_L + 2C_4C_LR_4R_Lg_m\right) + s\left(2C_1 + 2C_4R_4g_m + C_LR_4g_m + 2C_LR_Lg_m\right)}$$

9.19 X-INVALID-ORDER-19 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_4 R_4 g_m s + g_m}{C_1C_4C_LR_4s^3 + s^2\left(2C_1C_4 + C_1C_L + C_4C_LR_4g_m\right) + s\left(2C_4g_m + C_Lg_m\right)}$$

9.20 X-INVALID-ORDER-20 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{C_4 R_4 R_L g_m s + R_L g_m}{C_1C_4C_LR_4R_Ls^3 + g_m + s^2\left(C_1C_4R_4 + 2C_1C_4R_L + C_1C_LR_L + C_4C_LR_4R_Lg_m\right) + s\left(C_1 + C_4R_4g_m + 2C_4R_Lg_m + C_LR_Lg_m\right)}$$

9.21 X-INVALID-ORDER-21 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_4 C_L R_4 R_L g_m s^2 + g_m + s\left(C_4R_4g_m + C_LR_Lg_m\right)}{s^3\left(C_1C_4C_LR_4 + 2C_1C_4C_LR_L\right) + s^2\left(2C_1C_4 + C_1C_L + C_4C_LR_4g_m + 2C_4C_LR_Lg_m\right) + s\left(2C_4g_m + C_Lg_m\right)}$$

9.22 X-INVALID-ORDER-22 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ R_4,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1 R_4 R_L g_m}{R_1R_4g_m + 2R_1R_Lg_m + R_4 + 2R_L + s\left(C_1R_1R_4 + 2C_1R_1R_L\right)}$$

9.23 X-INVALID-ORDER-23 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m}{s^2\left(2C_1C_4R_1 + C_1C_LR_1\right) + s\left(2C_4R_1g_m + 2C_4 + C_LR_1g_m + C_L\right)}$$

9.24 X-INVALID-ORDER-24 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_1 R_L g_m s + R_1 g_m}{2C_1C_4C_LR_1R_Ls^3 + s^2\left(2C_1C_4R_1 + C_1C_LR_1 + 2C_4C_LR_1R_Lg_m + 2C_4C_LR_L\right) + s\left(2C_4R_1g_m + 2C_4 + C_LR_1g_m + C_L\right)}$$

9.25 X-INVALID-ORDER-25 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_1 R_4 R_L g_m s + R_1 R_4 g_m}{2C_1C_4C_LR_1R_4R_Ls^3 + 2R_1g_m + s^2\left(2C_1C_4R_1R_4 + C_1C_LR_1R_4 + 2C_1C_LR_1R_L + 2C_4C_LR_1R_4R_Lg_m + 2C_4C_LR_4R_L\right) + s\left(2C_1R_1 + 2C_4R_1R_4g_m + 2C_4R_4 + C_LR_1R_4g_m + 2C_LR_1R_Lg_m + C_LR_4 + 2C_LR_L\right) + 2}$$

9.26 X-INVALID-ORDER-26 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_4 R_1 R_4 g_m s + R_1 g_m}{C_1C_4C_LR_1R_4s^3 + s^2\left(2C_1C_4R_1 + C_1C_LR_1 + C_4C_LR_1R_4g_m + C_4C_LR_4\right) + s\left(2C_4R_1g_m + 2C_4 + C_LR_1g_m + C_L\right)}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4s},\ \infty,\ \frac{R_L}{C_LR_Ls+1}\right)$

$$H(s) = \frac{C_4R_1R_4R_Lg_ms + R_1R_Lg_m}{C_1C_4C_LR_1R_4R_Ls^3 + R_1g_m + s^2\left(C_1C_4R_1R_4 + 2C_1C_4R_1R_L + C_1C_LR_1R_L + C_4C_LR_1R_4R_Lg_m + C_4C_LR_4R_L\right) + s\left(C_1R_1 + C_4R_1R_4g_m + 2C_4R_1R_Lg_m + C_4R_4 + 2C_4R_L + C_LR_1R_Lg_m + C_LR_L\right) + 1}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4s},\ \infty,\ R_L + \frac{1}{C_Ls}\right)$

$$H(s) = \frac{C_4C_LR_1R_4R_Lg_ms^2 + R_1g_m + s\left(C_4R_1R_4g_m + C_LR_1R_Lg_m\right)}{s^3\left(C_1C_4C_LR_1R_4 + 2C_1C_4C_LR_1R_L\right) + s^2\left(2C_1C_4R_1 + C_1C_LR_1 + C_4C_LR_1R_4g_m + 2C_4C_LR_1R_Lg_m + C_4C_LR_4 + 2C_4C_LR_L\right) + s\left(2C_4R_1g_m + 2C_4 + C_LR_1g_m + C_L\right)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ \infty,\ R_4,\ \infty,\ R_L\right)$

$$H(s) = \frac{C_1R_1R_4R_Lg_ms + R_4R_Lg_m}{R_4g_m + 2R_Lg_m + s\left(C_1R_1R_4g_m + 2C_1R_1R_Lg_m + C_1R_4 + 2C_1R_L\right)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ \infty,\ \frac{1}{C_4s},\ \infty,\ \frac{1}{C_Ls}\right)$

$$H(s) = \frac{C_1R_1g_ms + g_m}{s^2\left(2C_1C_4R_1g_m + 2C_1C_4 + C_1C_LR_1g_m + C_1C_L\right) + s\left(2C_4g_m + C_Lg_m\right)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ \infty,\ \frac{1}{C_4s},\ \infty,\ R_L + \frac{1}{C_Ls}\right)$

$$H(s) = \frac{C_1C_LR_1R_Lg_ms^2 + g_m + s\left(C_1R_1g_m + C_LR_Lg_m\right)}{s^3\left(2C_1C_4C_LR_1R_Lg_m + 2C_1C_4C_LR_L\right) + s^2\left(2C_1C_4R_1g_m + 2C_1C_4 + C_1C_LR_1g_m + C_1C_L + 2C_4C_LR_Lg_m\right) + s\left(2C_4g_m + C_Lg_m\right)}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ \infty,\ \frac{R_4}{C_4R_4s+1},\ \infty,\ R_L + \frac{1}{C_Ls}\right)$

$$H(s) = \frac{C_1C_LR_1R_4R_Lg_ms^2 + R_4g_m + s\left(C_1R_1R_4g_m + C_LR_4R_Lg_m\right)}{2g_m + s^3\left(2C_1C_4C_LR_1R_4R_Lg_m + 2C_1C_4C_LR_4R_L\right) + s^2\left(2C_1C_4R_1R_4g_m + 2C_1C_4R_4 + C_1C_LR_1R_4g_m + 2C_1C_LR_1R_Lg_m + C_1C_LR_4 + 2C_1C_LR_L + 2C_4C_LR_4R_Lg_m\right) + s\left(2C_1R_1g_m + 2C_1 + 2C_4R_4g_m + C_LR_4g_m + 2C_LR_Lg_m\right)}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4s},\ \infty,\ \frac{1}{C_Ls}\right)$

$$H(s) = \frac{C_1C_4R_1R_4g_ms^2 + g_m + s\left(C_1R_1g_m + C_4R_4g_m\right)}{s^3\left(C_1C_4C_LR_1R_4g_m + C_1C_4C_LR_4\right) + s^2\left(2C_1C_4R_1g_m + 2C_1C_4 + C_1C_LR_1g_m + C_1C_L + C_4C_LR_4g_m\right) + s\left(2C_4g_m + C_Lg_m\right)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4s},\ \infty,\ \frac{R_L}{C_LR_Ls+1}\right)$

$$H(s) = \frac{C_1C_4R_1R_4R_Lg_ms^2 + R_Lg_m + s\left(C_1R_1R_Lg_m + C_4R_4R_Lg_m\right)}{g_m + s^3\left(C_1C_4C_LR_1R_4R_Lg_m + C_1C_4C_LR_4R_L\right) + s^2\left(C_1C_4R_1R_4g_m + 2C_1C_4R_1R_Lg_m + C_1C_4R_4 + 2C_1C_4R_L + C_1C_LR_1R_Lg_m + C_1C_LR_L + C_4C_LR_4R_Lg_m\right) + s\left(C_1R_1g_m + C_1 + C_4R_4g_m + 2C_4R_Lg_m + C_LR_Lg_m\right)}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4s},\ \infty,\ R_L + \frac{1}{C_Ls}\right)$

$$H(s) = \frac{C_1C_4C_LR_1R_4R_Lg_ms^3 + g_m + s^2\left(C_1C_4R_1R_4g_m + C_1C_LR_1R_Lg_m + C_4C_LR_4R_Lg_m\right) + s\left(C_1R_1g_m + C_4R_4g_m + C_LR_Lg_m\right)}{s^3\left(C_1C_4C_LR_1R_4g_m + 2C_1C_4C_LR_1R_Lg_m + C_1C_4C_LR_4 + 2C_1C_4C_LR_L\right) + s^2\left(2C_1C_4R_1g_m + 2C_1C_4 + C_1C_LR_1g_m + C_1C_L + C_4C_LR_4g_m + 2C_4C_LR_Lg_m\right) + s\left(2C_4g_m + C_Lg_m\right)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ R_4,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_1 C_L R_1 R_4 R_L g_m s^2 + R_4 g_m + s\left(C_1 R_1 R_4 g_m + C_L R_4 R_L g_m\right)}{2 g_m + s^2\left(C_1 C_L R_1 R_4 g_m + 2 C_1 C_L R_1 R_L g_m + C_1 C_L R_4 + 2 C_1 C_L R_L\right) + s\left(2 C_1 R_1 g_m + 2 C_1 + C_L R_4 g_m + 2 C_L R_L g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L} R_1 R_4 g_m \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + 2\sqrt{2}\sqrt{C_1}\sqrt{C_L} R_1 R_L g_m \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + \sqrt{2}\sqrt{C_1}\sqrt{C_L} R_4 \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + 2\sqrt{2}\sqrt{C_1}\sqrt{C_L} R_L \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}}}{2 C_1 R_1 g_m + 2 C_1 + C_L R_4 g_m + 2 C_L R_L g_m}$

wo: $\sqrt{2}\sqrt{\frac{g_m}{C_1 C_L R_1 R_4 g_m + 2 C_1 C_L R_1 R_L g_m + C_1 C_L R_4 + 2 C_1 C_L R_L}}$

bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{C_1 C_L R_1 R_4 g_m + 2 C_1 C_L R_1 R_L g_m + C_1 C_L R_4 + 2 C_1 C_L R_L}}\left(2 C_1 R_1 g_m + 2 C_1 + C_L R_4 g_m + 2 C_L R_L g_m\right)}{\sqrt{2}\sqrt{C_1}\sqrt{C_L} R_1 R_4 g_m \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + 2\sqrt{2}\sqrt{C_1}\sqrt{C_L} R_1 R_L g_m \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + \sqrt{2}\sqrt{C_1}\sqrt{C_L} R_4 \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + 2\sqrt{2}\sqrt{C_1}\sqrt{C_L} R_L \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}}}$

K-LP: $\frac{R_4}{2}$

K-HP: $\frac{R_1 R_4 R_L g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}$

K-BP: $\frac{C_1 R_1 R_4 g_m + C_L R_4 R_L g_m}{2 C_1 R_1 g_m + 2 C_1 + C_L R_4 g_m + 2 C_L R_L g_m}$

Qz: None

Wz: $\frac{1}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_L}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ R_L\right)$

$$H(s) = \frac{C_1 C_4 R_1 R_4 R_L g_m s^2 + R_L g_m + s\left(C_1 R_1 R_L g_m + C_4 R_4 R_L g_m\right)}{g_m + s^2\left(C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_1 R_L g_m + C_1 C_4 R_4 + 2 C_1 C_4 R_L\right) + s\left(C_1 R_1 g_m + C_1 + C_4 R_4 g_m + 2 C_4 R_L g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_4} R_1 R_4 g_m \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + 2\sqrt{C_1}\sqrt{C_4} R_1 R_L g_m \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + \sqrt{C_1}\sqrt{C_4} R_4 \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + 2\sqrt{C_1}\sqrt{C_4} R_L \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}}}{C_1 R_1 g_m + C_1 + C_4 R_4 g_m + 2 C_4 R_L g_m}$

wo: $\sqrt{\frac{g_m}{C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_1 R_L g_m + C_1 C_4 R_4 + 2 C_1 C_4 R_L}}$

bandwidth: $\frac{\sqrt{\frac{g_m}{C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_1 R_L g_m + C_1 C_4 R_4 + 2 C_1 C_4 R_L}}\left(C_1 R_1 g_m + C_1 + C_4 R_4 g_m + 2 C_4 R_L g_m\right)}{\sqrt{C_1}\sqrt{C_4} R_1 R_4 g_m \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + 2\sqrt{C_1}\sqrt{C_4} R_1 R_L g_m \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + \sqrt{C_1}\sqrt{C_4} R_4 \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}} + 2\sqrt{C_1}\sqrt{C_4} R_L \sqrt{\frac{g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}}}$

K-LP: R_L

K-HP: $\frac{R_1 R_4 R_L g_m}{R_1 R_4 g_m + 2 R_1 R_L g_m + R_4 + 2 R_L}$

K-BP: $\frac{C_1 R_1 R_L g_m + C_4 R_4 R_L g_m}{C_1 R_1 g_m + C_1 + C_4 R_4 g_m + 2 C_4 R_L g_m}$

Qz: None

Wz: $\frac{1}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}}$

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